Fairness Without Utility Collapse: A Benchmark of Intervention–Protocol and Adaptive Weighting Strategies in Federated Learning

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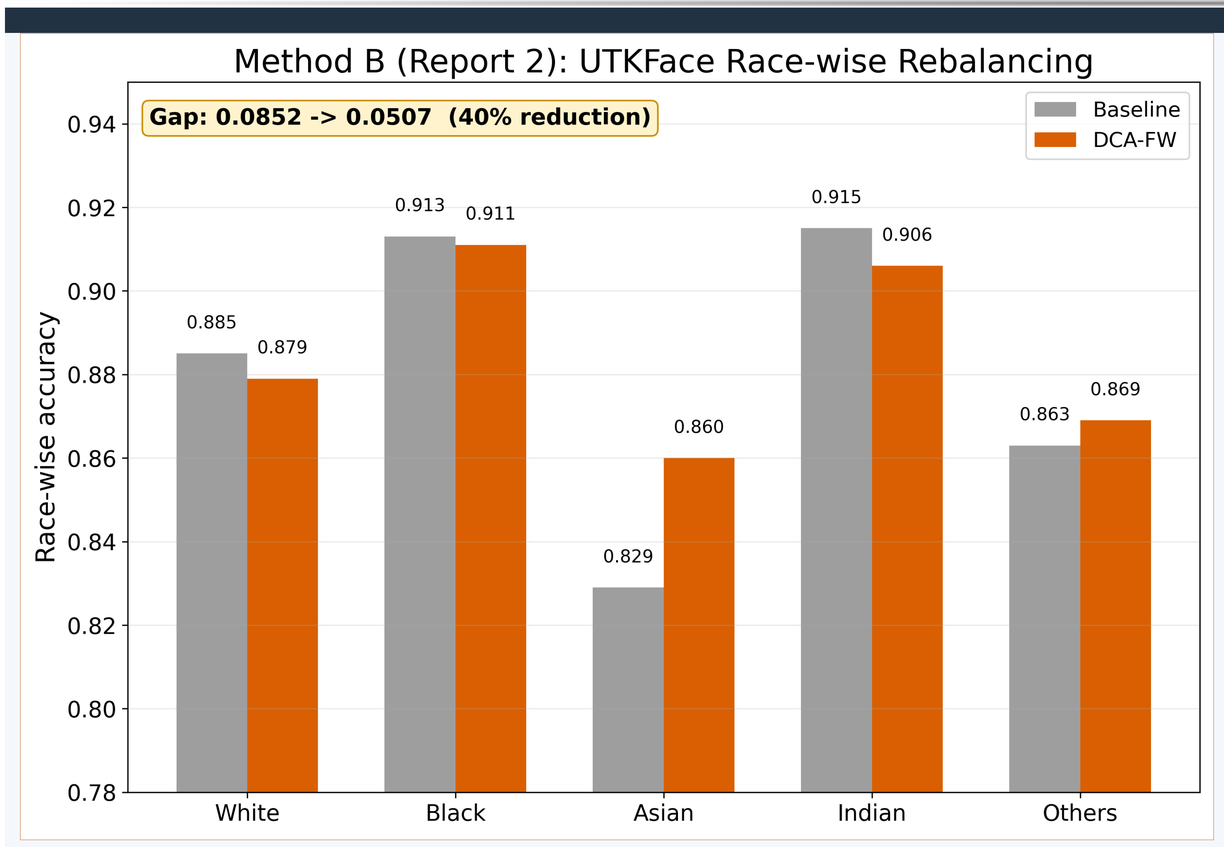
Take-Home Message

Fairness improvement in federated learning is achievable without mandatory utility collapse.

Tier 1: Combined Results Dashboard (Report 1 + Report 2)



Method A Pareto highlights (UTKFace, intervention+protocol)

Operating point	Acc	EO
A2+D2 (IID, AdamW)	0.9384	0.0426
A5+D3 (IID, AdamW)	0.9345	0.0379
A1+D2 (Dir-0.5, AdamW)	0.9128	0.0290

Method B FairFace validation (Report 2)

Setting	Gap	Acc
Baseline	0.0992	0.7723
DCA-FW run (40%)	0.0880	0.7790
DCA-FW run (50%)	0.0858	0.7695
DCA-FW best-gap run (60%)	0.0812	n/a

40%↓ UTKFace fairness gap (Report 2)

0.9384 Best Acc at A2+D2 (Report 1)

189 Total controlled FL runs (Report 1)

Tier 3: Shared Setup and Reading Rule

- Task: federated gender classification with race-aware fairness analysis.
- Common dataset: UTKFace; Report 2 also validates on FairFace.
- EO/DP (Report 1) and Fairness Gap (Report 2) are complementary fairness views.
- Interpret fairness gains only at non-degenerate utility levels.

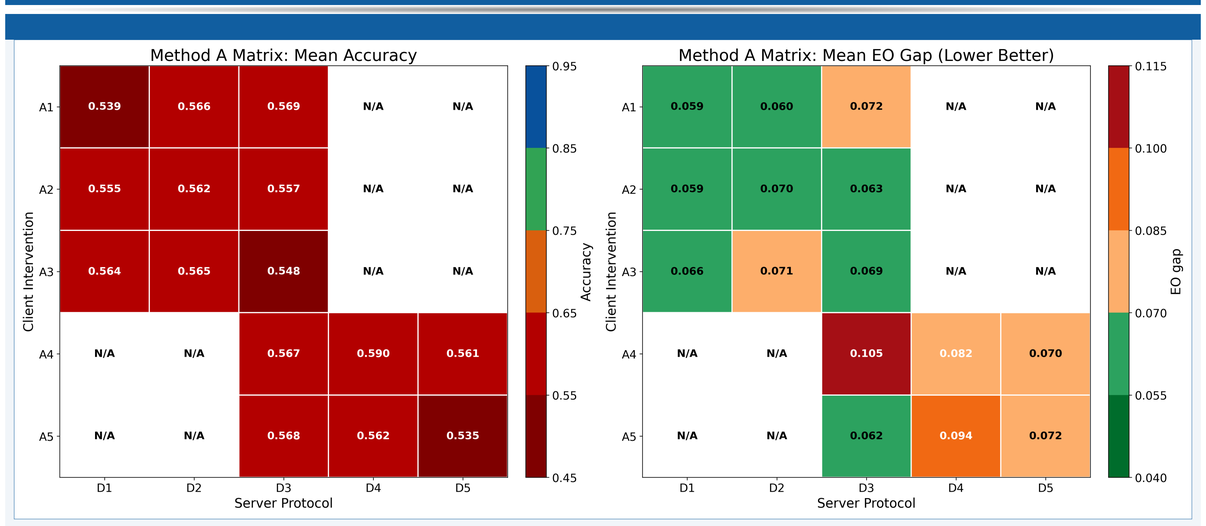
Method B Mechanism (Report 2): DCA-FW

Client-specific fairness weights are updated from observed race-wise performance, and then used to reweight local loss in the next round:

$$w_{k,r}^{(t+1)} = \frac{1}{a_{k,r}^{(t)} + \epsilon}, \qquad \mathcal{L}_k^{(t+1)} = \sum_r w_{k,r}^{(t+1)} \mathcal{L}_{k,r}$$

This report-level mechanism is what drives the

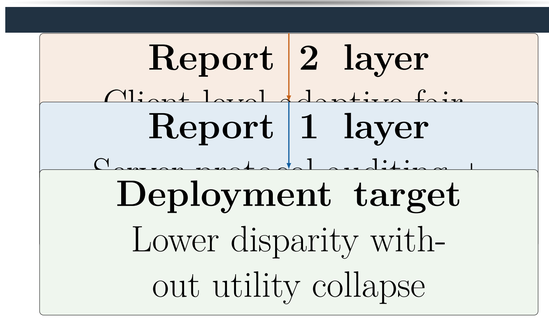
Tier 2: Figure 1 – High-Contrast Intervention–Protocol Matrix (Report 1)



Matrix cells report mean values across split × optimizer; unsupported pairings are explicitly marked as n/a.

Tier 2: Figure 2 – Pareto Frontier with Best Operating Point (Report 1)

Cross-Method Integration Pipeline



Tier 3: Limitations and Next Step

- Report 1 uses one seed for the 189-run profile.
- Report 2 includes one FairFace best-gap run with missing accuracy log.
- Next step: unified multi-seed benchmark where DCA-FW and protocol auditing are co-evaluated in one training stack.